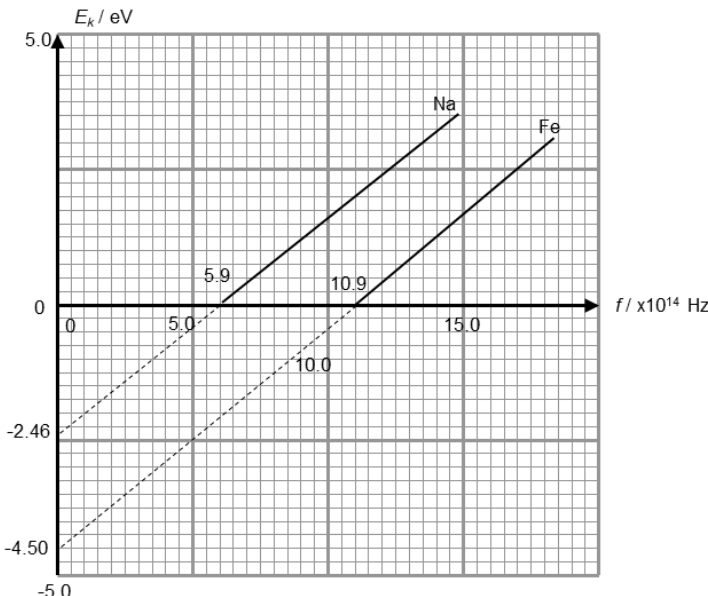


Q5	
(a)(i)	It is a <u>quantum</u> of <u>electromagnetic radiation</u> energy.
	The <u>energy of a photon</u> E is hf , where h is the Planck's constant and f is the frequency of the electromagnetic radiation.

(a)(ii)	$E = h \frac{c}{\lambda}$ $= 6.63 \times 10^{-34} \times \frac{3.00 \times 10^8}{390 \times 10^{-9}}$ $= 5.10 \times 10^{-19} \text{ J}$ $= \frac{5.10 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 3.188 \text{ eV}$ $= 3.19 \text{ eV (3 s.f.) (shown)}$
(a)(iii)1	 Parallel lines drawn Correct labeling on x-axis (5.9,10.9) Correct labeling on y-axis (-2.46,-4.50)
(a)(iii)2	Gradient – no change because the gradient is the Planck's constant and not dependent on the intensity of light Vertical intercept – no change because the vertical intercept is work function is dependent on the metal and not dependent on the intensity of light
(b)(i)	13.6 eV Electron energy level difference between lowest energy state to 0 eV.
(b)(ii)	The highest energy level is 0 eV. <u>OR</u> The electrostatic force between the electron and nucleus is attractive. <u>OR</u> Positive work done by external force to ionise the hydrogen atom.
(b)(iii)	Only photons with energy equal to the energy difference between the energy levels will be absorbed. The electrons in the lower energy state will absorb the photons and get excited to a higher energy level. Photons not absorbed will continue to travel to screen. Excited electrons will return to lower energy level by emitting photons in all directions, resulting in dark lines at these wavelengths as the photons is much smaller in numbers compared to the other unabsorbed photons.

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